

Month 4: End of Season (Heuvelink extract)

Purpose: end-of-season material from the academic reference on tomato physiology, for the Month 4 brainstorm. Source: Ep Heuvelink (ed.), *Tomatoes*, 2nd edition, CABI Crop Production Science in Horticulture 27, 2018 (`process/sources/Heuvelink2018_Tomatoes_CABI_2ndEd.pdf`, text in `process/sources/heuvelink-raw.txt`, split in `process/sources/heuvelink-chapters/`). Chapters 3, 4, 5, 7, 9 and 10. Status: DO NOT PUBLISH. Brainstorm source only. Dropped per `brainstorm-spec.md`: CO2 enrichment strategies, supplemental lighting and LEDs, closed/semi-closed greenhouses, stone wool and recirculation systems, grafting and transplant production, processing tomatoes, mechanical harvest, packing-line and controlled-atmosphere handling, breeding and genetics.

Topping (heading off)

- The growing point is removed 5 to 8 weeks before the anticipated crop termination date; a week later all remaining flowers are removed. (Ch9, p295)
- An individual fruit needs 6 to 9 weeks from anthesis to harvest, so flowers or small fruits present after topping never reach maturity. (Ch9, p295)
- Some growers leave a few shoots at the top, or do not top at all, to shade the top fruit and keep transpiration up, which lowers cracking and russetting risk. (Ch9, p295)
- Time from flower to ripe fruit is set by temperature, not by fruit load, light, CO2, humidity, plant age or density. The countback is a temperature countback. (Ch3, p77-78)
- Rule of thumb for one cultivar: about 940 degree-days above a 4°C base from anthesis to harvest-ripe. Fruit development rate rises linearly with temperature from 14°C to 26°C. (Ch3, p77)
- Temperature sensitivity is highest in the first weeks after anthesis and drops later, but close to maturity extra warmth still speeds ripening. (Ch3, p77-78)

Fall light and what it can pay for

- Yield per unit of light is lower in autumn than in spring, for two reasons: a lower leaf area index and lower CO2 in the house. The same light buys less in fall. (Ch4, p107)
- The one-percent rule: 1% less light gives 0.7 to 1% less production. Dirty glazing and shade structures cost yield directly. (Ch4, p107; Ch9, p281)
- The light effect is stronger on fresh fruit weight than on plant dry weight, and the relative effect of light shrinks at low temperature and low CO2. (Ch4, p107)
- Low light delays inflorescence initiation, cuts flower development, and causes flower abortion. Late clusters set poorly in fall for a physical reason, not a management failure. (Ch4, p107)
- Under low light, fruit set can be temporarily improved by removing some young leaves, which compete with fruit for assimilates. (Ch4, p107-108)
- White plastic on the ground reflects 50 to 80% of PAR against 10 to 20% for bare soil, and lifts crop photosynthesis at least 7% at LAI 3. (Ch4, p94)
- Light interception saturates: at LAI 3, 90% of incident light is already intercepted. Leaves beyond that add shade, not sugar. (Ch4, p93)

Leaf count and deleafing

- Standard practice: all leaves below the bottom truss removed, and 12 to 18 fully expanded leaves left on the stem, set by the grower. (Ch9, p294-295)
- Leaves are removed up to just above the truss about to be harvested, or pruned to a target leaf-to-fruit ratio. (Ch4, p94)
- Leaf pruning alone raises the share of dry matter going to fruit from 66% to 74%, but yield barely moves, because a lower leaf area index intercepts less light. (Ch4, p104)
- Leaf pruning combined with delaying removal of the old leaves, so leaf area index stays up, was predicted to raise yield 13%. Deleaf hard at the head, slow at the bottom. (Ch4, p104-105)
- Above a leaf-to-fruit ratio of 1, extra leaves add nothing to fruit growth, but they do raise fruit dry matter and sugar content. (Ch4, p105)
- Deleafing timing is a three-way trade-off: removing leaves takes out immature whiteflies, but also destroys the parasitoids developing inside them, while leaving the pile in the house spreads disease. (Ch9, p295)
- Where parasitized pupae are known to be present, leaves are pulled but left piled inside for a few days so the wasps can emerge. (Ch9, p295)
- Trimming old leaves before the wasps emerge sets back parasitoid populations; an adapted trimming pattern is the standard advice. (Ch7, p216)
- Over-pruning burns fruit: direct sun can raise exposed fruit tissue 10°C or more above air temperature. (Ch5, p165)

Fall temperature

- Fruit-production stage targets: 20 to 22°C day, 17 to 19°C night. (Ch9, p301)
- No growth below 12°C, and long exposure there causes chilling injury to the plant. (Ch4, p115)
- Temperature integration: hold the average rather than fixed day and night numbers. A crop tolerates 3°C below standard for 6 days if the next 6 days run 3°C above, and even 6°C swings if the 7-day average holds. Energy savings of 10 to 15%. (Ch9, p301)
- Since development follows the 24 h mean, warm days let night temperatures fall to save fuel, as long as the mean holds. (Ch9, p301)
- Average fruit size falls as temperature rises from 17°C to 26°C: warmth makes more trusses and faster fruit, so the same assimilates spread over more fruit. Cooler fall running gives bigger last fruit. (Ch4, p115)
- Optimum for vegetative growth is 18 to 25°C; truss appearance rate rises nearly linearly from 17°C to 27°C; optimal anthesis temperature is 18 to 20°C. (Ch4, p115)
- Between 17 and 23°C the vegetative-to-generative ratio does not depend on the average temperature. Steering happens through the day/night spread, not the mean. (Ch4, p115)
- Maintenance respiration roughly doubles per 10°C rise. Warmth without light is burned sugar. (Ch4, p114-115)
- A cold night after a warm day (25°C/9°C against 25°C/15°C for 9 days) cut CO₂ assimilation capacity. Cold nights cost more than the night itself. (Ch4, p116)
- Grow pipes, heat pipes hung inside the canopy on flexible hoses, are moved vertically to put heat exactly where fruit is ripening. (Ch9, p302)

Steering the balance at the end

- A balanced plant: stem about 1 cm thick 15 cm below the tip, dark green leaves, closely spaced large clusters that set well. Thicker means too vegetative and poor set; thinner means carbohydrate starvation. (Ch9, p298)
- Generative levers: higher EC, less frequent and lower-volume irrigation, high temperature, low relative humidity, lower nitrogen or a high potassium-to-nitrogen ratio. (Ch9, p298)
- Thick head reading: raise day temperature 1 to 2°C in peak light, and widen the day/night spread. The bigger the spread, the stronger the generative signal. (Ch9, p299)
- Thin head reading: bring day and night closer together, and cut the 24 h average in low radiation, which is exactly the late-autumn case. Target a 10 to 12 mm head. (Ch9, p299)
- Head tight, leaves not unfolding until late in the day: raise the 24 h average by warming between midnight and sunrise. Curl should be out between 11am and 4pm. (Ch9, p299)
- Pale yellow flowers in the morning read as low humidity deficit, under 2 g/m³. Fix by raising VPD to 3 to 7 g/m³ early, venting briefly to dump moisture then heating. Flowering rate should hold at 0.8 to 1.0 truss per week. (Ch9, p299)
- Sticky flowers whose sepals do not roll back come from a too-humid morning. Activate the plant with minimum pipe heat and a cracked vent. Left alone they make poor fruit. (Ch9, p300)
- Long straight kinked trusses come from low light plus high temperature: lower the 24 h temperature by lowering the day, and keep an active climate near 3 g/m³. (Ch9, p299)
- Truss hooks or braces stop heavy trusses from kinking or pulling off the stem; braces go on before the fruit develops, while the stem is still flexible. (Ch9, p295)
- Long cropping cycles swing from too vegetative early to too generative late, with too little growth and excessive fruit load at the end. (Ch9, p298-301)
- Total assimilate demand averages about twice the supply over a season, so fruit and leaves are always competing. (Ch4, p106)
- Fruit yield has an optimum against fruit number per truss: more fruit raises the fruit share but shrinks leaf area and total biomass; fewer fruit sizes up the ones left. (Ch4, p91)
- Fruit pruning is how a target average fruit weight is set; typical cluster practice is four to six fruit per truss. (Ch4, p91; Ch9, p295)
- Misshapen and undersized fruit at the end of a cluster are always removed: they never make marketable size and they shrink the rest of the truss. (Ch9, p295)
- Side shoots come from every axil and are removed weekly. Extra shoots are only kept when light is high against fruit load. (Ch9, p294)
- Fruit under source limitation reach their full potential growth rate about 2 weeks after the limitation is lifted by fruit pruning. Thinning pays back with a two-week delay. (Ch3, p78)

Humidity, condensation and calcium

- Diseases spread fast when VPD stays at or below 0.2 kPa for long stretches, and spore germination climbs on wet leaf surfaces. (Ch9, p306)
- The classic fall trap: warm sunny days load the air with vapour, then leaf and glazing surfaces cool to the dewpoint at night and drip lands on the leaves. (Ch9, p306)

- Fruit heats far slower than air at sunrise, so condensation lands on fruit first. If no condensation forms on the fruit, none forms on the leaves. Watch fruit temperature, not air temperature. (Ch9, p307)
- Raising air movement to 1 m/s cuts condensation: it thins the boundary layer, keeps leaf temperature closer to air temperature, and keeps surfaces above the dewpoint. (Ch9, p307)
- Horizontal airflow fans run continuously above the canopy, pushing air one way down one side and back the other, which also evens out temperature. (Ch9, p307)
- When humidity is too high but temperature is fine, the fix is heating and venting together: heater on to hold temperature, vents open. (Ch9, p306)
- Sustained high relative humidity (low VPD) cut leaf area through calcium deficiency, cut final yield and cut mean fruit weight. (Ch9, p306)
- VPD above 1.0 kPa is considered potentially stressful; if plants transpire more than roots can supply, fruit gets blossom-end rot and stomata close. (Ch9, p306)
- Roll-up and double-poly houses with rounded roofs are the worst case for condensate dripping onto leaves, because the arch cannot collect and channel it. (Ch9, p282, p306)
- Energy-saving construction, especially double poly, raises house humidity as a side effect. (Ch9, p305)

Fall disease pressure

- Grey mould is an opportunist: it colonizes tissue wounded during production or by insects, and it is typically associated with mature plants carrying a dense canopy in cooler weather. It does not need moisture beyond what sits inside the canopy. (Ch7, p229-230)
- It starts on old senescent tissue, moves into younger leaves, then petioles, then stems. Stem infections girdle the plant and wilt everything above. (Ch7, p230)
- Stem infections start at pruning wounds, at graft unions, and where stems rub on string or stakes. (Ch7, p230)
- After humid spells, infected tissue releases a visible cloud of spores when it is moved. Handle infected material gently, and take it out. (Ch7, p230)
- Senescent flower petals are an infection route: the fungus colonizes the petal, then the developing fruit, and petals lodged in stem-petiole junctions start stem infections. (Ch7, p230)
- Ghost spot, a necrotic fleck with a white halo appearing as fruit ripens, marks where the fungus penetrated young fruit and died. (Ch7, p230)
- Infected fruit that is not rogued turns into large postharvest losses in shipment. (Ch7, p230)
- Removing and destroying senescent and dead plant tissue is the primary control. Spores travel long distances on wind, so hygiene is about the local load. (Ch7, p230-231)
- Liming acidic soil limits the disease by improving plant calcium. (Ch7, p231)
- Leaf mould needs relative humidity above 85% and 22 to 24°C, though it develops anywhere from 4 to 32°C. It starts on older leaves and can move onto fruit as a black dry rot at the stem end. (Ch7, p233)
- Leaf mould control: remove and destroy diseased debris, hold humidity below 85%, and prune and stake for airflow through the canopy. The fungus survives as a saprophyte on leaf tissue in the soil. (Ch7, p233)
- Powdery mildew: the two *Oidium* species grow on the leaf surface, so mycelium and spores are visible; *Leveillula taurica* grows inside the leaf, shows only pale angular vein-delimited spots up top, and sporulates through the stomata below only above 95% humidity. (Ch7, p233-234)

- Old crop residues are a main reservoir of whitefly-borne viruses. Destroying them is an important control. (Ch7, p216)
- Low greenhouse temperatures, driven by energy saving, restrict Encarsia and Eretmocerus efficacy. Running cold to save fuel costs whitefly control. (Ch7, p216)
- Predatory mites do not work on tomato: they get trapped in the sticky stem hairs. Whitefly control rests on mirid bugs and parasitoids. (Ch7, p216)
- Airborne disease including Botrytis is minimized by climate management, which is what makes biological control workable in a greenhouse. (Ch7, p208-209)
- Standard annual cycle: plants in the house 11 months, in production about 9, with roughly a month between crops for clean-up and pest and disease control. (Ch9, p285)

Fall irrigation

- Fertigation should start a few hours after sunrise and end 1 to 2 h before sunset, which cuts disease as well as russetting and cracking. (Ch9, p298)
- Frequent small irrigations beat infrequent large ones for holding the root zone in range. (Ch9, p284)
- Late-stage feed target for the fifth truss to termination: nitrogen 150, potassium 200, calcium 150 ppm, EC 1.8 dS/m, against 120/150/150 and EC 1.5 earlier. Potassium and nitrogen climb toward the end. (Ch9, p290)
- Moderate water or salt stress in the mid to late stages of fruit development raises sugars, acids, aroma, vitamins and lycopene with little yield loss. Late season is when a controlled squeeze buys flavour. (Ch5, p154)
- Fruit dry matter content rises linearly with EC while fruit size falls with it. The trade is size against taste. (Ch5, p154)

End-of-season fruit disorders

- Cracking happens when the inside expands faster than the skin. Susceptibility rises as fruit matures, especially once colour develops. Prevent with even watering, adequate calcium, and crack-resistant varieties. (Ch5, p163)
- Russetting is the fine-crack version: it does not invite much infection but it raises water loss and wrecks appearance. (Ch5, p163-164)
- Blossom-end rot is a local calcium shortage in the distal tissue during rapid fruit expansion. Conducive: high temperature, high light, high VPD, salinity, irregular watering, plant vigour, high nitrogen, root damage. (Ch5, p162)
- BER climbs sharply below 0.08% fruit calcium (dry weight) and is rare above 0.12%. Spraying calcium directly on young fruit is the recommended prevention. (Ch5, p163)
- Blotchy ripening, hard grey to yellow patches near the calyx, is fed by potassium or boron deficiency, high nitrogen, chilling, temperature above 30°C, low light or high soil moisture. Low light and chilling are both fall conditions. (Ch5, p164)
- Green shoulder is worsened by heat, high salinity and short potash and phosphate. (Ch5, p165)
- Sunscald: tissue above 30°C stops making red pigment, above 40°C the tissue dies, whitens and dries into a parchment patch. It follows sudden exposure, typically after over-pruning leaves. (Ch5, p165)
- Silvering shows up in short-day crops and after cold shocks. (Ch5, p166)
- Puffiness comes from bad pollination or seed development under too-low or too-high temperature, high nitrogen or low light, and hits the distal fruit of the cluster first. (Ch5, p164-165)

- Fruit temperatures above 26°C during growth cut lycopene and beta-carotene. Low growing temperatures above 10°C favour ascorbic acid, phenolics and carotenoids. (Ch5, p154)
- Fruit irradiance itself, not leaf photosynthesis or sugar supply, drives vitamin C accumulation. Fruit that never sees light never gets there. (Ch5, p154)
- Firmness drops in summer heat and high VPD; high EC and high magnesium raise it, while high calcium supply lowers it. (Ch5, p155)

Finishing fruit off the vine

- Ripening speed off the vine, days from each stage to table-ripe: (Ch10, p333)

Stage at pick	12.5°C	15°C	17.5°C	20°C	22.5°C	25°C
Mature-green	18	15	12	10	8	7
Breaker	16	13	10	8	6	5
Turning	13	10	8	6	4	3
Pink	10	8	6	4	3	2

- Mature-green fruit ripens best at 15 to 20°C. Above 25°C it comes out soft and poorly coloured. (Ch10, p330)
- Ripening is uneven and abnormal below 10°C. (Ch10, p332)
- Above 30°C lycopene synthesis stops and fruit turns orange instead of red. (Ch10, p332)
- Storage windows: mature-green 2 weeks at 12.5 to 15°C, ripening fruit 1 week at 10 to 12.5°C, fully ripe 3 to 5 days at 7 to 10°C. (Ch10, p330)
- Chilling injury starts below about 12°C: slow abnormal ripening, more decay, less acidity, faster water loss, surface pitting. (Ch10, p330-331)
- Fruit picked hot is more chilling sensitive than fruit picked cool, and fruit grown in a hot climate is more sensitive than fruit grown cool. (Ch10, p331)
- Fruit chilled on the plant before harvest carries that sensitivity into storage. Cold nights late in the season cost shelf life after the pick. (Ch10, p331)
- Riper pink fruit is more chilling tolerant than mature-green, because chilling injury mostly shows up as failed later ripening. (Ch10, p332)
- Once a fruit hits breaker, it makes its own ethylene and finishes on its own. External ethylene adds nothing after that point. (Ch10, p335)
- Fruit picked below about 90% of final size never reaches good quality, even though it will eventually turn red and soften. (Ch10, p327)
- Ripening rooms are held at 85 to 95% relative humidity to stop ripening fruit from losing water to its own respiratory heat. (Ch10, p326)
- Respiration rises 20 to 50% going from mature-green to red, so stacked fruit needs airflow to carry the heat away. (Ch10, p325-326)
- Greenhouse fruit is picked riper than field fruit, so it bruises easier and keeps shorter. Handle the last picks less, not more. (Ch10, p315)

- Greenhouse fruit is picked two or three times a week, more often in warm weather when fruit develops faster. Fall picks space out. (Ch10, p315)
- Fruit in the sun without canopy shade heats enough to damage tissue and spoil later ripening. Cover the bins. (Ch10, p319; Ch5, p165)
- Ethylene applied to the whole plant before harvest is not used on multi-pick crops: it triggers flower drop, leaf drop and general senescence. Paste on a single truss is the workaround. (Ch10, p329)
- Immersing fruit deep in water pushes water and contaminants in through the stem scar. Wash water should be a few degrees warmer than the fruit. (Ch10, p320)
- Fruit sitting in water longer than 6 hours cracks and loses shelf life. (Ch10, p318)
- Gas exchange in a tomato happens almost entirely through the stem scar, so corking or wax there changes the whole internal atmosphere. (Ch10, p334)